UNITED INTERNATIONAL UNIVERSITY

V1.3 / 2026

URC

URRT

URRT ROBO CUP

SOFTWARE & AUTONOMOUS

01

MOBILITY

Rails slopes gravel and narrow paths

02

DEXTERITY

QR codes touch targets box smash and the sensor crate

03

AUTONOMY

Hands-off execution worth far more points

rescuerover.uiu.ac.bd • rescuerover@cair.uiu.ac.bd

#URRT • #URRT • #URRT • #URRT • #URRT • #URRT •

DOCUMENT SET

This is the **Software & Autonomous Rulebook**, part of the four-document URRT Robo Cup set:

- **General Rulebook** – format, milestones, arena, tasks, scoring including the Build Quality Score, run procedure, penalties, ranking.
- **Mechanical & Electrical Rulebook** – rover size, mechanical design and CAD, electrical and wiring rules, E-stop, power, communications, classes, safety.
- **Software & Autonomous Rulebook** – control modes, autonomy, sensor crate, dashboard.
- **Judgement Sheet** – inspection, score, dashboard and build quality sheets, milestone form, judge quick reference.

Where a rule lives in another document, this book points to it by name.

1. CONTROL MODES AND DECISIONS

1.1 Control Modes

The control mode used for a task sets the **control multiplier**. Autonomy is the pillar that is worth significantly more points.

Control mode	Definition	Control multiplier
Teleoperation	The operator drives the rover and triggers the task through the operator station. The rover may use driver aids (e.g. stabilization) but not decide the task outcome.	×1
Autonomous	After the start command the rover executes the task fully on its own. No human control input, no remote trigger. Detection, decision and action are all on board.	×4

Table 1. Control modes and control multipliers.

1.2 Autonomous Mode Rules

1.2.1 A task is scored as autonomous only if the team **declares** autonomy to the referee *before* the task starts. The declaration is logged on the score sheet.

1.2.2 From the start command until the end of the autonomous task, the operator may not issue control commands, move joysticks, send waypoints or give any hint through any channel. The operator may only press the E-stop.

1.2.3 Telemetry and video may be displayed at the operator station for the audience, but the rover must not rely on any off-board computation or command.

1.2.4 Any autonomous sequence that depends on pre-programmed distances, timings or positions is permitted; however, the rover must still use its sensors to complete the task (e.g. read the QR code in the lane, not a memorized value).

1.2.5 An autonomous task that is aborted by the team through any input is scored as teleoperation from that moment (see Table 2).

1.3 Decision Table

Teams constantly make decisions during a run. The table below summarizes how each decision affects the control multiplier of the affected task. The class multiplier is unaffected by these decisions.

Team decision	Control mult.	Penalty	Notes
Declare autonomous; task completed with no input	×4	None	Full autonomy bonus.
Declare autonomous; operator aborts and completes it manually	×1	None	Task falls back to teleop; the autonomy bonus is lost.
Declare autonomous; E-stop or reset needed	×0	Reset rules	No points for that attempt; see the General Rulebook.
Do not declare; complete the task by teleop	×1	None	Standard scoring.
Declare autonomous but cannot complete	×0	None	Task remains available to retry by teleop if time remains.
Attempt an autonomous task a second time after a first failure	×2	None	Only one retry per task; the second attempt is capped at ×2.
Skip the task	–	None	No points; lane can still be unlocked by other tasks.

Table 2. Effect of team decisions on the control multiplier.

AUTONOMY AND THE UNLOCK RULE

Autonomous and teleoperated points count the same toward the 3-point unlock threshold of a lane, because the threshold uses base points only.

1.4 Eligibility of Tasks for Autonomous Scoring

Task	Autonomous eligible	Max control mult.
Continuous lane traverse (Mobility)	YES	×4
QR decode and ping (Sensor Crate)	YES	×4
Touch target (Dexterity)	YES	×4
Color/shape detection	YES	×4
Sonar proximity check	YES	×4
Motion tracking	YES	×4
Box smash (Lane 1)	YES	×4
Optional: payload carry and leaning obstacle	YES	×4
Two-way audio acuity	NO	×1

Table 3. Which tasks accept the autonomous multiplier.

2. SENSOR CRATE

The Sensor Crate is the set of sensing tasks that the rover's on-board sensors must complete. The tasks are distributed in the lanes as shown in the General Rulebook.

Sensor task	Description	Eligible mode
Two-way audio acuity	The rover plays an operator message through its speaker that the referee must understand; the operator must report the code word spoken by the referee through the rover's microphone.	Teleop
Visual QR decoding	The rover decodes a QR code on a panel using its camera. The QR code contains an IP address.	Teleop / Auto
Server ping	After decoding, the rover pings the IP from the QR code and the referee's server logs the echo request.	Teleop / Auto
Sonar proximity	The rover stops within a marked window in front of a wall using its sonar or ultrasonic sensor, or reports the distance within a tolerance.	Teleop / Auto
Colour and shape detection	The rover identifies the colour and shape of a card shown in the lane.	Teleop / Auto
Motion tracking	The rover detects a moving target and keeps it in the camera view for 5 seconds.	Teleop / Auto

Table 4. Sensor crate tasks.

2.1 QR Decode and Ping

This is the signature task of the championship: the rover reads a QR code with its camera and, using the decoded IP address, pings that address on the arena network.

2.1.1 The QR code contains the IPv4 address of the referee's ping server. The code changes between runs and between lanes.

2.1.2 Decoding the QR code earns the **decode** points. Pinging the decoded address earns the **ping** points. The ping points cannot be earned without a correct decode.

2.1.3 The ping must originate from the rover's own on-board computer. A ping sent from the operator laptop or any off-board device does not count.

2.1.4 In Autonomous Mode the decode and the ping must both be performed with no human input. The task is then scored with $\times$**4**; in teleop it is scored with $\times$**1** (the General Rulebook).

2.1.5 The referee's server log is the only valid proof of the ping. A single successful echo request in the log is sufficient.

2.2 Other Sensor Tasks

2.2.1 **Audio:** the referee reads a random four-digit code at the lane station. The rover relays the audio to the operator (listening, 2 base points), and the operator transmits a phrase back through the rover speaker, which the referee must be able to hear clearly (talking, 2 base points).

2.2.2 **Sonar:** the rover shall report or stop at a distance of 100 ± 20 mm from the marked wall.

2.2.3 **Colour/shape:** the referee shows one card out of a set of coloured shapes. The rover (or operator) reports the correct colour and shape.

2.2.4 **Motion:** a referee moves a target at slow walking speed in front of the rover. The rover must track it continuously for 5 seconds.

3. DASHBOARD CHALLENGE

3.1 Purpose and Overview

Every team builds a **dashboard**: a web interface through which the operator monitors the rover, switches between teleoperation and autonomous mode, drives the rover and sees all sensors, the camera and the emergency stop. The dashboard is assessed separately from the arena runs in a **Dashboard Review**, and its marks are added to the final ranking (the General Rulebook).

DASHBOARD IN ONE LINE

A dashboard **built on the rover's own circuit**, kept in a **GitHub repository**, that streams live data over a **socket**, exposes **correct endpoints**, and lets the operator **switch mode, control the robot, see every sensor and the camera, and trigger an E-stop**.

3.2 Dashboard Requirements

3.2.1 The dashboard must be built on and connected to the rover's own electronic circuit (on-board controller, sensors and camera). Simulated, mocked or pre-recorded data earns no marks in the functional categories of Table 7.

3.2.2 The repository must contain the circuit schematic or wiring diagram and the pin map of the rover circuit, so that the committee can see how the dashboard data is produced.

3.2.3 Each team must keep all dashboard code in a **GitHub repository** that is created before Milestone M1 and shared with the Team Mentor (public, or private with the Mentor invited). Code that is not in the repository is not assessed.

3.2.4 The dashboard must run in a standard browser on the operator laptop and work on the arena network without internet access.

3.2.5 The dashboard is the operator interface at the operator station. In Autonomous Mode all control inputs must be disabled and the dashboard must show the declared mode clearly, in line with Section 1.

3.2.6 The dashboard E-stop is an *additional* software E-stop. It never replaces the hardware E-stop required in the Mechanical & Electrical Rulebook and shall not be used as a reason to omit it.

3.2.7 Third-party libraries and frameworks are allowed if their licences permit; the dashboard logic, integration and design must be the team's own work.

3.3 Required Dashboard Functions

Function	Requirement
Mode switching	Operator can switch between Teleoperation and Autonomous mode from the dashboard; the active mode and the time of the last change are always visible.
Robot control	In Teleop the operator can drive the rover (and the flippers, where fitted) from the dashboard. Controls are locked in Autonomous Mode.
All sensors	Live values for every Sensor Crate sensor fitted: audio status, QR decoded text, ping result, sonar distance, colour/shape result and motion tracking status, each with a time stamp.
Camera	Live camera view inside the dashboard with a visible frame rate or latency indicator.
E-stop	A clearly visible E-stop control that stops all actuators and shows the E-stop state until reset.
Rover status	Connection state, battery level and current mode, updated live.

Table 5. Functions the dashboard must provide.

3.4 Interface Contract

The dashboard is assessed against the following interface. The organizers may publish the final endpoint list at the technical briefing; the list below is the default.

Method	Endpoint / event	Purpose
GET	`/api/status`	Connection, battery, current mode and E-stop state as JSON.
GET	`/api/sensors`	Latest time-stamped readings of all sensors as JSON.
POST	`/api/mode`	Switch mode with body `{"mode":"teleop"|"auto"}`.
POST	`/api/drive`	Send drive and flipper commands; rejected with an error while in Autonomous Mode.
POST	`/api/estop`	Engage the software E-stop; a reset endpoint releases it.
GET	`/api/camera/stream`	Live camera stream (MJPEG or equivalent).
WS	`/ws`	Socket connection pushing `sensors`, `status` and `estop` events to the dashboard.

Table 6. Default dashboard interface contract.

3.5 Dashboard Marking

The Dashboard Score has a maximum of 100 marks, plus an optional bonus.

Criterion	What is assessed	Max marks
Polling time	Time between live updates of sensor data in the dashboard (Table 8).	10
Correct endpoints	Endpoints exist, use the right methods and return correct, well-formed JSON.	10
Socket used	A working socket connection (e.g. WebSocket) pushes live data and status.	10
Intuitive UI/UX	Clear layout, readable status, obvious mode and E-stop controls, usable in a run.	10
Repo structure and commit quality	Organized repository, README, circuit files; meaningful, regular commits by team members.	10
Mode switch and robot control	Switching mode and controlling the rover from the dashboard with controls locked in Autonomous Mode.	20
All sensors	Every fitted Sensor Crate sensor shown live in the dashboard.	15
Camera	Live camera inside the dashboard.	10
E-stop	E-stop available and working inside the dashboard.	5
Total		**100**
Optional bonus	**Bad frontend design (judged separately, awarded at the committee's discretion)**	**up to +5**

Table 7. Dashboard marking scheme.

Median polling time of sensor data	Marks
$\leq$ **100 ms**	10
$\leq$ **250 ms**	8
$\leq$ **500 ms**	6
$\leq$ **1000 ms**	3
$>$ **1000 ms but working**	1
No live updates	0

Table 8. Polling time marks.

3.6 Dashboard Review

3.6.1 The Dashboard Review takes place as part of Milestone M4 and lasts 15 minutes per team. The committee measures the polling time live (median interval over 60 seconds), checks the endpoints, tests mode switching, control, sensors, camera and E-stop on the real rover, and walks through the user interface.

3.6.2 The repository is frozen at the M4 deadline. Commits after the freeze are not counted for repository marks, but fixes may be demonstrated.

3.6.3 Commit quality is judged on clear messages, regular small commits over the whole build and contributions from more than one team member. A single large upload shortly before the deadline receives low marks.

3.6.4 The **Dashboard Score** is the sum of all marks in Table 7, including any bonus, and is added once to the team's final total.

3.6.5 Teams are encouraged to use the dashboard as the operator interface during their runs, but arena scores do not depend on it.